Quantification: 535_z13c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	19
interior lacuna count	17
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 17)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2897.65	2801	2660.16	px^2
major_axis_length_px	lacuna shape	88.6456	94.2445	47.2848	px
minor_axis_length_px	lacuna shape	40.4127	38.3287	20.0816	px
aspect_ratio	lacuna shape	2.202	2.2844	0.7153	unitless
eccentricity	lacuna shape	0.8404	0.8991	0.1221	unitless
solidity	lacuna shape	0.7978	0.7829	0.0875	unitless
roots_count	headline	8.3529	7	5.7439	count
ring_length_r30_px	headline	258.941	234	168.691	px
ring_length_r60_px	reported	516.529	413	328.385	px
owned_length_px	ownership-dependent	604.466	391.836	561.317	px
edge_count	ownership-dependent	30.1765	18	29.2216	count
mean_edge_length_px	ownership-dependent	21.1983	19.5918	9.6981	px
perimeter_px	normalised	281.691	264.007	188.318	px
ring_area_r30_px2	normalised	9005.35	10205	4614.98	px^2
ring_area_r60_px2	normalised	18817.8	20860	8701.03	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9993	1	0.0028	unitless
ring_density_r30	normalised	0.0273885	0.0281138	0.00839756	px^-1
ring_density_r60	normalised	0.0263634	0.0255887	0.00812337	px^-1
roots_per_100px_perimeter	normalised	3.0027	2.7138	1.6279	per 100 px
ring_attached_length_r30_px	network v2	226.353	189	165.725	px
ring_attached_length_r60_px	network v2	400.059	318	332.75	px
ring_length_w_r30_px	network v2	292.059	262.066	191.972	px
ring_length_w_r60_px	network v2	583.511	460.948	374.299	px
sholl_crossings_r10	network v2	6.5882	7	4.6106	count
sholl_crossings_r20	network v2	6.3529	8	3.7071	count
sholl_crossings_r30	network v2	7.3529	6	4.4009	count
loop_count	loops, report only	1.3529	1	1.7657	count
loop_edge_share	loops, report only	0.1168	0.0882	0.126	share
ring_width_mean_r30_px	width	7.2215	7.4875	1.1937	px

Per field
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measure	value	units
analysed_area_px2	990845	px^2
total_skeleton_length_px	20810	px
canalicular_length_density_per_px	0.0210023	px^-1
junction_count	373	count
junction_density_per_px2	0.00037645	px^-2
lacuna_count	19	count

lacunae_per_px2	1.918e-05	px^-2
skeleton_component_count	340	count
mean_component_length_px	61.2059	px
median_component_length_px	20	px
field_length_density_w_per_px	0.0236467	px^-1
field_density_without_flagged_per_px	0.0204977	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	799	px
unattached_component_count	48	count
canalicular_length_density_attached_per_px	0.0210023	px^-1
field_length_density_with_unattached_per_px	0.0218087	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	27	count
loop_edge_share	0.081	share
edge_count	1087	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.3456	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	19426	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	3449	94.2445	51.4301	1.8325	0.838	0.8008	-1.3442	87.5868	746.098	False
2	1518	84.7212	28.0385	3.0216	0.9436	0.7609	0.4889	372.306	481.817	False
3	916	42.5998	32.8395	1.2972	0.637	0.7829	0.4554	434.044	377.415	False
4	6729	150.978	65.1171	2.3186	0.9022	0.7721	-0.3046	496.001	291.95	False
5	3320	105.162	44.1209	2.3835	0.9077	0.7721	-1.4464	464.635	577.918	False
6	3645	102.737	49.6538	2.0691	0.8754	0.8697	0.3597	580.157	513.438	False
7	2801	98.9222	38.3287	2.5809	0.9219	0.8173	0.524	580.027	151.776	False
8	9054	174.453	85.8111	2.033	0.8707	0.6586	0.1605	649.475	679.118	False
9	1209	70.3083	25.9548	2.7089	0.9294	0.8103	-1	737.055	638.021	False
10	3256	105.273	45.222	2.3279	0.903	0.7485	-1.2191	649.232	325.526	False
11	524	31.117	22.6042	1.3766	0.6872	0.8748	-1.3629	760.422	888.611	False
12	597	63.166	16.0382	3.9385	0.9672	0.6958	-1.4137	764.35	853.412	False
13	543	31.4075	23.6295	1.3292	0.6588	0.9282	0.087	780.65	298.357	False
14	455	27.4749	22.5328	1.2193	0.5722	0.9362	1.1384	784.457	318.629	False
15	7705	171.941	75.2669	2.2844	0.8991	0.7155	-1.3369	837.132	645.403	False
16	2929	114.614	39.6482	2.8908	0.9383	0.6816	1.0561	854.515	315.604	False
17	610	37.8566	20.7792	1.8218	0.8359	0.937	-0.7458	884.313	871.107	False
18	2259	90.1772	47.5611	1.896	0.8496	0.6829	0.6436	985.683	487.498	True
19	6212	244.279	36.0383	6.7783	0.9891	0.7991	-1.5538	1007.31	703.743	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	9	393	1014	1592.07	82	19.4154	7.1937
2	7	181	367	123.953	11	11.2685	5.2487
3	11	234	413	769.997	33	23.3332	7.4626
4	17	485	939	1529.41	84	18.2072	8.2352
5	7	261	359	225.024	12	18.752	6.6633
6	3	204	417	202.64	7	28.9485	8.5039
7	12	263	534	499.031	24	20.793	7.9127
8	13	518	1025	1538.63	68	22.6269	7.8684
9	5	72	91	211.409	4	52.8523	5.8056
10	13	390	559	403.948	18	22.4416	7.669
11	1	97	294	196.196	8	24.5245	7.4875
12	4	74	229	174.74	11	15.8855	6.9188
13	2	86	206	123.669	7	17.667	8.7709
14	3	75	138	391.836	20	19.5918	8.2757
15	17	514	1026	1157.16	72	16.0717	7.6634
16	17	461	858	1130.38	51	22.1643	6.8189
17	1	94	312	5.8284	1	5.8284	4.2665
18	6	270	480	255.208	17	15.0122	8.2074
19	8	288	673	548.86	31	17.7052	8.3581

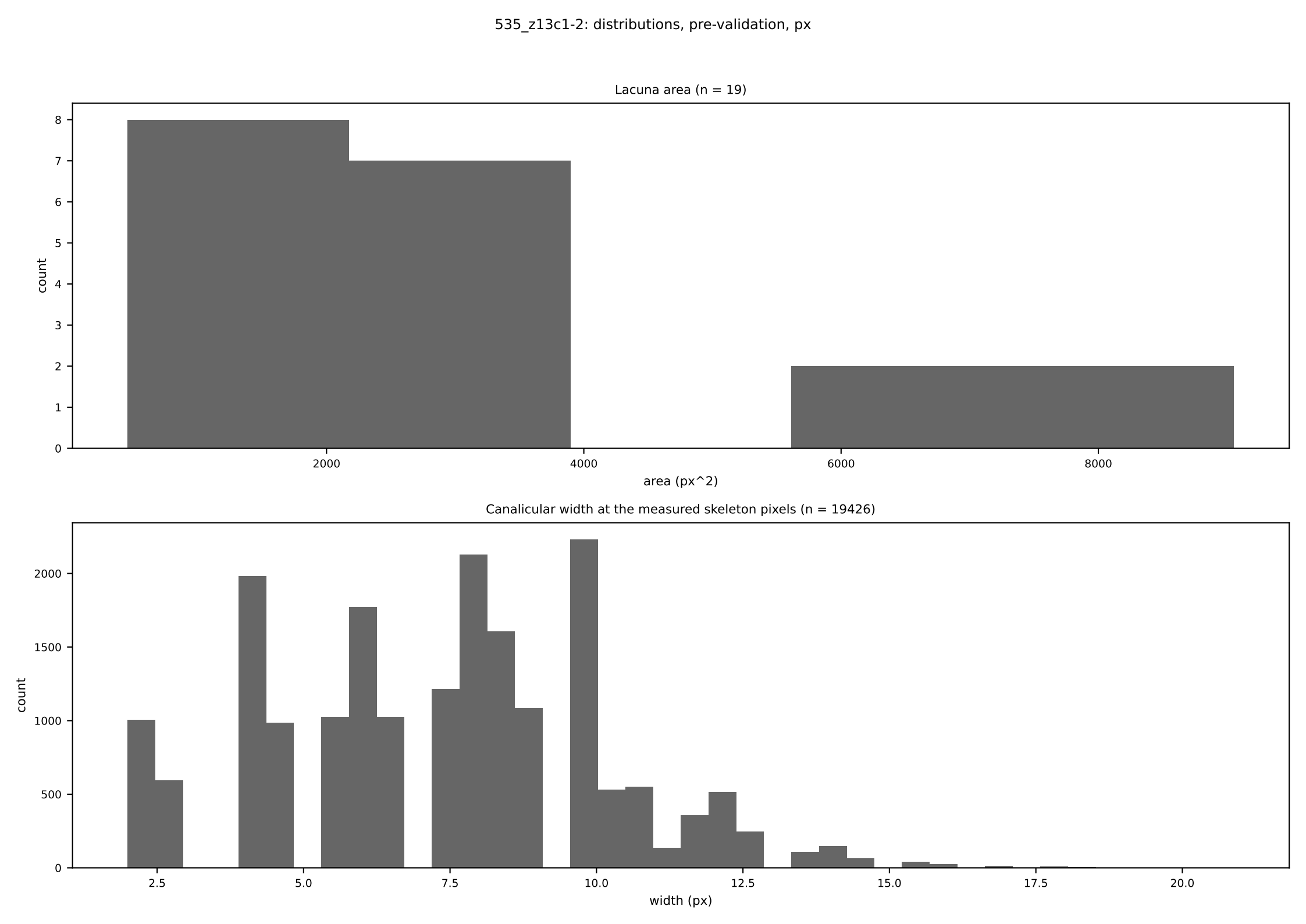
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	311.848	10999	27029	1	0.9885	0.0357305	0.0375153	2.886
2	215.622	8573	18773	1	1	0.0211128	0.0195494	3.2464
3	145.782	5773	12342	1	1	0.0405335	0.033463	7.5455
4	433.103	13694	24741	1	1	0.035417	0.0379532	3.9252
5	326.563	11088	20860	1	1	0.023539	0.01721	2.1435
6	270.936	10229	21506	1	1	0.0199433	0.0193899	1.1073
7	264.007	10205	23754	1	1	0.0257717	0.0224804	4.5453
8	731.429	16860	32093	1	1	0.0307236	0.0319384	1.7773
9	194.894	5275	8641	1	1	0.0136493	0.0105312	2.5655
10	328.877	11820	23768	1	1	0.0329949	0.023519	3.9528
11	99.6335	3693	10047	1	1	0.0262659	0.0292625	1.0037
12	147.397	4813	11557	1	1	0.015375	0.0198148	2.7138
13	96.0833	3059	6126	1	1	0.0281138	0.0336272	2.0815
14	85.734	2578	5393	1	1	0.0290923	0.0255887	3.4992
15	655.553	17714	33766	1	1	0.0290166	0.0303856	2.5932
16	380.475	11026	23669	1	1	0.0418103	0.03625	4.4681
17	100.811	5692	15838	1	1	0.0165144	0.0196995	0.992
18	275.764	8026	17251	0.7864	0.7193	0.0336407	0.0278245	2.1758
19	597.973	11032	21934	0.5965	0.5686	0.0261059	0.030683	1.3379

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	385	868	442.09	1162.02	10	9	12
2	99	121	202.894	411.789	4	6	7
3	224	371	262.066	470.789	7	6	6
4	452	894	551.931	1065.34	13	10	14
5	189	210	288.309	393.492	6	12	6
6	186	318	226.095	460.948	7	8	5
7	249	430	303.078	603.115	7	8	8
8	499	933	585.902	1161.42	14	12	14
9	68	87	83.2548	106.397	1	2	1

10	316	361	447.475	633.428	8	8	11
11	84	192	114.296	337.907	3	3	5
12	74	142	79.8701	248.782	1	1	1
13	63	118	94.1838	238.338	2	1	5
14	47	72	80.2843	148.083	1	3	2
15	487	886	576.718	1156.79	15	9	13
16	394	766	524.961	973.051	10	8	11
17	32	32	101.598	348.007	3	2	4
18	254	354	304.522	540.174	7	9	7
19	243	422	315.865	750.027	7	10	8


535_z13c1-2: distributions, pre-validation, px
Lacuna area (n = 19)
count
area (px^2)
0
1
2
3
4
5
6
7
8
2000
4000
6000
8000
Canalicular width at the measured skeleton pixels (n = 19426)
count
width (px)
0
500
1000
1500
2000
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0